Langlin Huang

Email: h.langlin@wustl.edu | **LinkedIn**: langlin-huang | **Tel**: 206-643-8598 | **Homepage**: shrango.github.io

Education

Washington University in St. Louis (WashU) *St. Louis, U.S.*
Ph.D. Candidate in Computer Science – GPA: 4.0/4.0 Aug. 2024 – Present

Institute of Computing Technology, Chinese Academy of Sciences *Beijing, China*
M.E. in Computer Technology – GPA: 3.75/4.0, Outstanding Dissertation Sep. 2021 – Jun. 2024

University of International Business and Economics (UIBE) *Beijing, China*
B.E. in Data Science and Big Data Technology – GPA: 3.74/4.0 (Rank: 2/147) Sep. 2017 – Jun. 2021

Research Interests

- **Post-Training for LLM Reasoning**: RL with verifiable rewards, on-policy distillation, process reward models, self-evolution
- **LLM Agents**: agent harness optimization, long-horizon context management, agentic RL
- **Efficient Inference**: speculative decoding, efficient test-time scaling, collaborative decoding

Publications

(denotes equal contribution)*

Post-Training (RL and On-Policy Distillation)

- **On the Off-Policy Teacher in On-Policy Distillation**
  Langlin Huang*, Hao Liu*, Mononito Goswami, Xinyu Li, Prithwish Jana, Nikos Kanakaris, Patrick Blöbaum, Purak Jain
  NeurIPS 2026 Workshops: MATH-AI and Pre-to-Post [Paper] [Project]
- **Nonsense Helps: Prompt Space Perturbation Broadens Reasoning Exploration**
  Langlin Huang, Chengsong Huang, Jinyuan Li, Donghong Cai, Yuyi Yang, Jiaxin Huang
  EMNLP 2026 [Paper] [Code]
- **G-Zero: Self-Play for Open-Ended Generation from Zero Data**
  Chengsong Huang, Haolin Liu, Tong Zheng, Runpeng Dai, **Langlin Huang**, Jinyuan Li, Zongxia Li, Zhepei Wei, Yu Meng, Jiaxin Huang
  NeurIPS 2026 Spotlight [Paper] [Code]
- **Training Data Efficiency in Multimodal Process Reward Models**
  Jinyuan Li, Chengsong Huang, **Langlin Huang**, Shaoyang Xu, Haolin Liu, Wenxuan Zhang, Jiaxin Huang
  ICML 2026 [Paper] [Code]
- **Questioning the Questions: Sustaining Self-Evolution in Reasoning Models**
  Jinyuan Li, Chengsong Huang, **Langlin Huang**, Donghong Cai, Shiping Gao, Yuyi Yang, Jiaxin Huang
  Preprint 2026 [Paper] [Code] [Project]
- **Process Rewards with Learned Reliability**
  Jinyuan Li, **Langlin Huang**, Chengsong Huang, Shaoyang Xu, Donghong Cai, Yuyi Yang, Wenxuan Zhang, Jiaxin Huang
  Preprint 2026 [Paper] [Code]

Agent

- **MILO: Automated Harness Discovery via Orchestrated Multi-Agent Evolution**
  Prithwish Jana, Mononito Goswami, Hao Liu, Xinyu Li, **Langlin Huang**, Zhehui Huang, Zhishen Huang, Patrick Blöbaum, Anoop Deoras, Purak Jain, Nikos Kanakaris, Sahika Genc
  Preprint 2026 [Paper] [Project]
- **Hermes: Learning Contextual Reasoning Unlocks Test-Time Scaling**
  Xinyu Li, Mononito Goswami, Hao Liu, Nikos Kanakaris, **Langlin Huang**, Prithwish Jana, Patrick Blöbaum, Purak Jain
  NeurIPS 2026 Workshop: Long-Context Foundation Models [Paper]

Efficient LLM

- **SPEED: Specialized Position Experts for Efficient Speculative Decoding**
  Langlin Huang, Chengsong Huang, Jixuan Leng, Di Huang, Jiaxin Huang
  COLM 2026 [Paper] [Code]
- **CaTS: Calibrated Test-Time Scaling for Efficient LLM Reasoning**
  Chengsong Huang, **Langlin Huang**, Jixuan Leng, Jiacheng Liu, Jiaxin Huang
  ICLR 2026 [Paper] [Code]
- **RelayLLM: Efficient Reasoning via Collaborative Decoding**
  Chengsong Huang, Tong Zheng, **Langlin Huang**, Jinyuan Li, Haolin Liu, Jiaxin Huang
  EMNLP 2026 [Paper] [Code]

Multilingual

- **MoCE: Adaptive Mixture of Contextualization Experts for Byte-based Neural Machine Translation**
  Langlin Huang, Mengyu Bu, Yang Feng
  NAACL 2025 Oral [Paper] [Code]
- **Integrating Multi-scale Contextualized Information for Byte-based Neural Machine Translation**
  Langlin Huang, Yang Feng
  ACL 2024 Findings [Paper] [Code]
- **Enhancing Neural Machine Translation with Semantic Units**
  Langlin Huang, Shuhao Gu, Zhuocheng Zhang, Yang Feng
  EMNLP 2023 Findings [Paper] [Code]
- **BayLing: Bridging Cross-lingual Alignment and Instruction Following through Interactive Translation for Large Language Models**
  Shaolei Zhang, Qingkai Fang, Zhuocheng Zhang, Zhengrui Ma, Yan Zhou, **Langlin Huang**, Mengyu Bu, Shangtong Gui, Yunji Chen, Xilin Chen, Yang Feng
  Technical Report [Paper] [Code]

Others

- **Divide, Reweight, and Conquer: A Logit Arithmetic Approach for In-Context Learning**
  Chengsong Huang, **Langlin Huang**, Jiaxin Huang
  EACL 2026 Oral [Paper] [Code]
- **CrossWordBench: Evaluating the Reasoning Capabilities of LLMs and LVLMs with Controllable Puzzle Generation**
  Jixuan Leng, Chengsong Huang, **Langlin Huang**, Bill Yuchen Lin, William W. Cohen, Haohan Wang, Jiaxin Huang
  COLM 2025 [Paper] [Code] [Dataset]
- **Automatic Construction of a Depression-Domain Lexicon Based on Microblogs: Text Mining Study**
  Genghao Li, Bing Li, **Langlin Huang**, Sibing Hou
  JMIR Medical Informatics, Vol. 8, Jun. 2020 [Paper]

Industry Experience

AWS Agentic AI DevOps, Amazon

Applied Scientist Intern May 2026 – Sep. 2026

Identified the *off-policy teacher issue* in On-Policy Distillation (OPD). Proposed updating the teacher policy conditioned on the student prefixes, improving the student's performance on math reasoning and code generation.

- We found that even a stronger model performs badly when continuing from a worse model-generated prefix, showing its inadequacy as a teacher.
- We proposed a strategy named Student-COnditioned Update of the Teacher (SCOUT), which effectively alleviates the mismatch between teacher and student models.
- We empirically showed that sparse teacher-side updates are sufficient to mitigate the off-policy teacher issue and improve final performance.

Search Relevance Team, ByteDance

Algorithm Engineer Jan. 2024 – Apr. 2024

Proposed an adapter module to bridge the relevance model used in production with an LLM. The adapter transformed representations from the traditional model into LLM-compatible hidden states, enabling:

- Joint utilization of pretrained traditional models and powerful LLM reasoning capabilities.
- Robust handling of out-of-vocabulary and multilingual inputs.
- A representation model plus LLM framework extensible to user-isolated data or multimodal input scenarios.

Community Services

- **Top Reviewer**: NeurIPS 2025
- **Reviewer**: NeurIPS, ICML, ICLR, COLM, ACL, ACL Rolling Review, TALLIP